

Ahmed Ikram | Data Engineering

Dundee, Scotland | +44 7421 996048 | ahmedikram30@gmail.com | [LinkedIn](#) | [GitHub](#) | [Online Portfolio](#)

Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree, graduating June 2027. Builds data pipelines across three different paradigms - event streaming on AWS, batch lakehouse ETL on Azure, real-time OLAP streaming on GCP - matching the architecture to the traffic pattern each time. A self-hosted ClickHouse platform ingests 58.9M+ real-time events with zero data loss across 8 chaos-engineering injections. Seeking a graduate Data Engineering role.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree. Expected Graduation: **June 2027**

TECHNICAL SKILLS

Orchestration & Transformation: Apache Airflow · Apache Kafka · dbt · Apache Spark / PySpark · Pydantic Data Contracts

Streaming & Real-Time: ClickHouse · Materialized Views (SummingMergeTree) · Server-Sent Events · Idempotent Consumers · At-Least-Once Delivery

Modeling & Warehousing: Kimball Star Schema · SCD Type 1 · Delta Lake · BigQuery · Databricks (DLT, Unity Catalog)

Data Quality & Observability: Great Expectations · dbt Tests · Grafana · Prometheus · Azure Monitor · Cloud Monitoring · StatsD · Freshness SLAs & Column-Level Lineage

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, IAM, CloudFront, SageMaker) · Azure (Databricks, Azure SQL, Data Factory, ADLS) · Google Cloud (Compute Engine, BigQuery, GCS, Secret Manager, Workload Identity Federation) · Terraform · Docker · GitHub Actions · Linux

Languages & Databases: Python · SQL · PostgreSQL · MongoDB · MySQL · SQLite · Redis · Alembic

PROJECTS

W3C Web Logs ETL Pipeline | Apache Airflow · Databricks · dbt · PySpark · Azure SQL · Delta Lake · Terraform · Power BI · DAX · Power Automate · Grafana · Prometheus

- Databricks DLT medallion pipeline (Bronze → Silver) ingests 93 IIS log files (155,380 rows, 0 dropped) via 7 DLT quality gates; consolidated MaxMind GeoIP struct UDF (3.5× faster than 7 separate UDFs), achieving 99.99% known-country coverage across 153,377 enriched.
- 16 dbt models build a Kimball-style star schema across 9 dimension tables (SCD Type 1 on hashed natural keys), compiling against both PostgreSQL and T-SQL from one codebase via 18 compatibility macros; 121 dbt tests gate every model before Power BI, catching 213 duplicate rows via 12-component dedup key.
- Unity Catalog gives column-level lineage and centralised governance across bronze/silver/gold; 2 Airflow DAGs orchestrate the Databricks pipeline with zero processing logic in DAG files and 30+ Azure resources deploy via one Terraform apply with full OIDC federation configurations with a rollback workflow.
- A freshness probe and 3 Grafana dashboards (23 panels, 10 alert rules) back a 7-page Power BI dashboard (155K requests, 88 countries); 613 tests across 5 tiers, including dual-dialect dbt CI.

WikiStream Real-Time Streaming Analytics Platform | Python · ClickHouse · Grafana · BigQuery · Terraform · Google Cloud · Pydantic · Great Expectations

- Async consumer streams Wikipedia's public EventStreams feed (hand-rolled SSE parser) into a self-hosted ClickHouse cluster - 58.9M+ events ingested in 3 days, sustaining 203.5 events/sec average with real-world peaks of 2,719/sec; Pydantic schema-on-write validation rejects malformed events to a dead-letter table before they ever reach disk.
- At-least-once delivery via a durable cursor and a dedup ring buffer proven with a live SIGKILL-mid-insert test (zero loss and zero duplicates); an 8-injection chaos battery fired and cleared every rule across Grafana and Cloud Monitoring.
- 3 materialized views cut dashboard query latency 15x at p50 (13.2x p99) and rows scanned ~200x, verified at 2.08x real peak load (5,655 events/sec sustained, zero drops); hourly expost BigQuery keeps a 30-day-plus warehouse tier, parity-checked by exact-sum comparison every run.
- Great Expectations gates freshness, nulls, cardinality and schema drift hourly; GCP-provisioned via Terraform with WIF, itemised FinOps tracking held to \$41.65/month run-rate, validated by 143 tests at 99.4% coverage.

- Data engineering lead on a 7-person Agile team for NCR Atleos; independently replaced a direct-DB generator with an event-driven Kafka KRaft pipeline processing 2.5M+ events per day at 100+ msg/sec across 7 sources, routing malformed records to a dead-letter table without halting ingestion.
- At-least-once delivery with idempotent deduplication (hybrid Redis SET+TTL, LRU fallback) holds effectively-once behaviour across consumer restarts; backed by a Redis Stream DLQ with exponential backoff before manual triage
- PostgreSQL data lake on JSONB payloads - new sources only need a new parser, zero migrations; 14 composite indexes sustain sub-100ms queries past 2M+ rows under a pooled, retrying connection layer
- MLflow (RDS backend + S3 artifacts) tracks every training run with champion-alias promotion; full AWS estate (10 Terraform modules, 118 resources: multi-AZ VPC, ECS Fargate, RDS, S3, CloudFront, 6 least-privilege IAM roles) validated by 1,438 tests catching data loss under 50 concurrent writes

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant *September 2023 - Present*

- Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker *June 2021 - September 2023*

- Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure

CERTIFICATES

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| • <u>AWS Academy Graduate - Machine Learning Foundations</u> | <i>Feb 2025</i> |
| • <u>AWS Academy Graduate - Cloud Foundations</u> | <i>Mar 2024</i> |
| • <u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u> | <i>Oct 2024</i> |